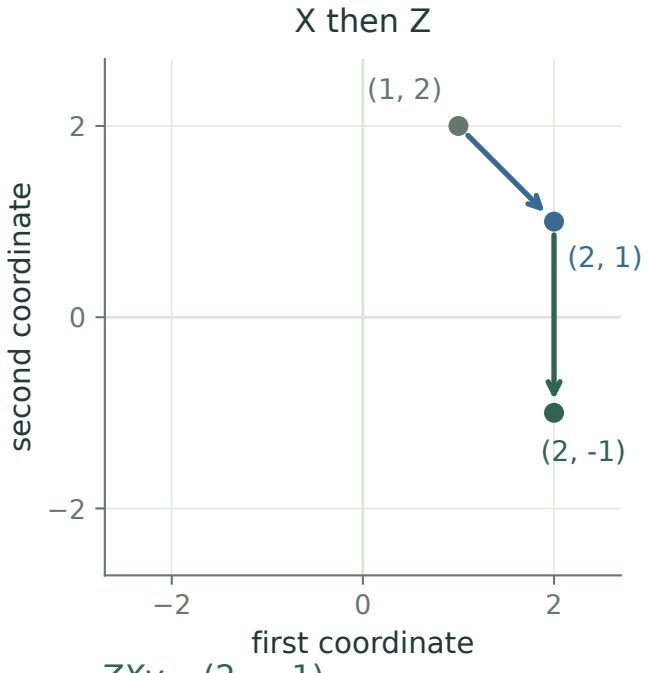
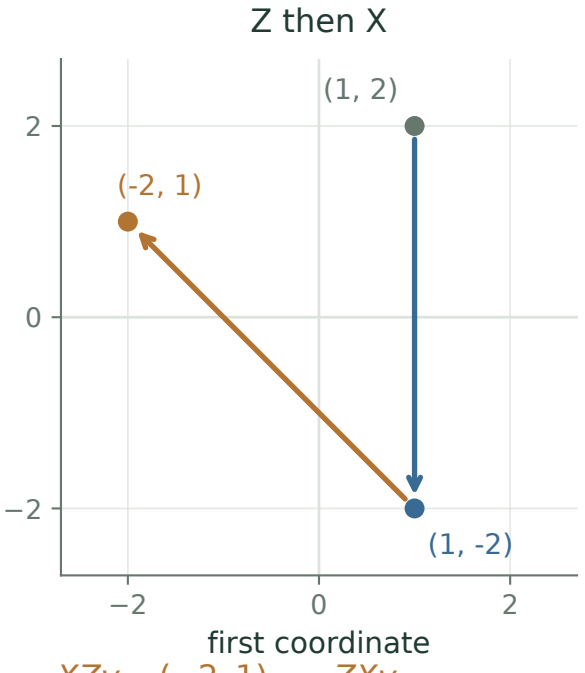
What survives when the order is reversed?

Apply the same two matrices to v = (1, 2).



$ZXv = (2, -1)$



$XZv = (-2, 1) = -ZXv$

$X(x, y) = (y, x)$; $Z(x, y) = (x, -y)$. The two-bit sum is the same; the sign differs.